Aalto University
School of Arts, Design
and Architecture

Check point no.4
Topic: Distributed Mixture-of-Experts (MoE) systems

Phuong Ngo Pitkänen and Amirreza Jafariandehkordi

Topic

Routing-aware expert: Rebalancing for MoE training

MoE models are widely used for scalable LLM training

Performance strongly depends on:

- distributed execution
- expert placement
- communication efficiency

Real HPC clusters (CSC LUMI) provide an ideal platform for testing.

The topic combines:

- practical engineering
- performance analysis
- novel research contribution

Our Goal:
Improve efficiency of distributed MoE training beyond existing studies.

Phase 1 and Phase 2 Overview

Phase 1 – Benchmarking and Setup

Deploy MoE training on CSC/LUMI

Use existing frameworks:

- DeepSpeed **
- Megatron

Establish reproducible baseline results

Understand bottlenecks and limitations

Phase 2 – Innovation

Build routing-aware expert rebalancing

Integrate with training workflow

Evaluate improvement over baselines

Key idea:
First measure → then improve.

Timeline

Week 6-7

Environment setup on CSC/LUMI

Repository and shared documentation

Literature review

Week 8-9

Basic MoE training pipeline

Multi-GPU experiments

Initial benchmarks

Week 10-11

Rebalancing techniques implementation

Week 12-13

Experiments and evaluation

Week 14

Final report writing and deliverables packaged

Week 15

Demo and presentation

Weekly Progress Overview

Planned this week

Environment setup on CSC / LUMI

Repository and shared documentation

Literature review

Actual progress

Repository and documentation prepared

Literature review continued

Started building Phase 1 MoE training baseline

Initial MoE Training Baseline

What we implemented:

- Small baseline MoE model
- Basic expert routing mechanism
- Synthetic data for controlled training
- Configurable training pipeline

What is does:

- Creates starting point for experiments
- Enables controlled performance testing later

Current State of the Pipeline

What we did:

- Training workflow
- Modular structure for model and training
- Reproducible configuration

Current limitations:

- Still a basic prototype
- Single-GPU only
- Not yet optimized or fully robust

LUMI Issues

Main issue this week

- LUMI frequently unavailable
- Last two days completely inaccessible
- Could not run or test any experiments

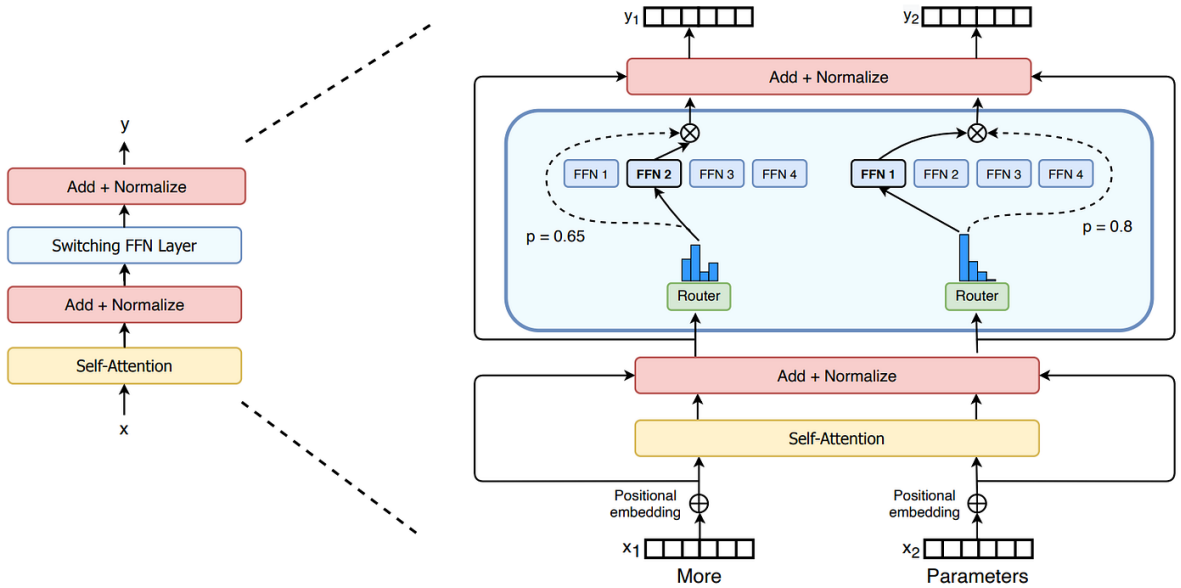
We attempted SSH key configuration

But we were still unable to connect

Next Steps

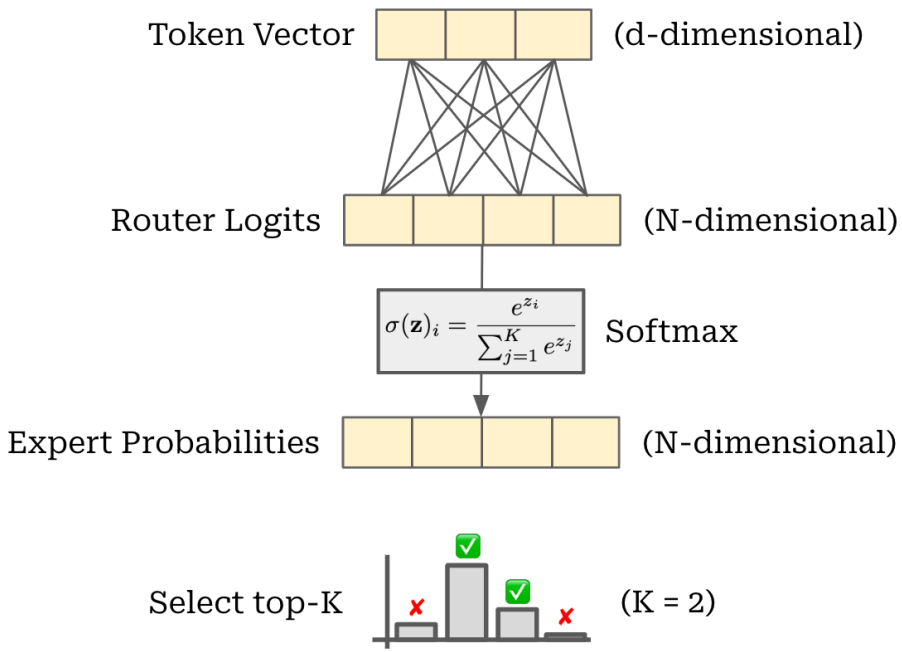
Technical parts:	Once LUMI works:
• Make training pipeline more robust • Prepare distributed execution • Run multi-GPU experiments • Collect initial benchmarks	• Validate baseline performance • Begin performance analysis

Fundamentals of MoEs



Source: Fedus, William, Barret Zoph, and Noam Shazeer. "Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity." Journal of Machine Learning Research 23.120 (2022): 1-39.

Routing algorithms: selecting experts



Source: Shazeer, Noam, et al. "Outrageously large neural networks: The sparsely-gated mixture-of-experts layer." arXiv preprint arXiv:1701.06538 (2017).

"Routing collapse":

“The gating network tends to converge to a state where it always produces large weights for the same few experts. This imbalance is self-reinforcing, as the favored experts are trained more rapidly and thus are selected even more by the gating network.”

A!

Kiitos
aalto.fi